

Credit Card Default Auto-ML Report

🕒 2026-05-08 12:37:04 UTC

🏆 Best: **catboost** roc_auc = 0.7790

📊 학습 24000행

테스트 6000행

Feature 197개

1 모델 비교 (holdout 기준)

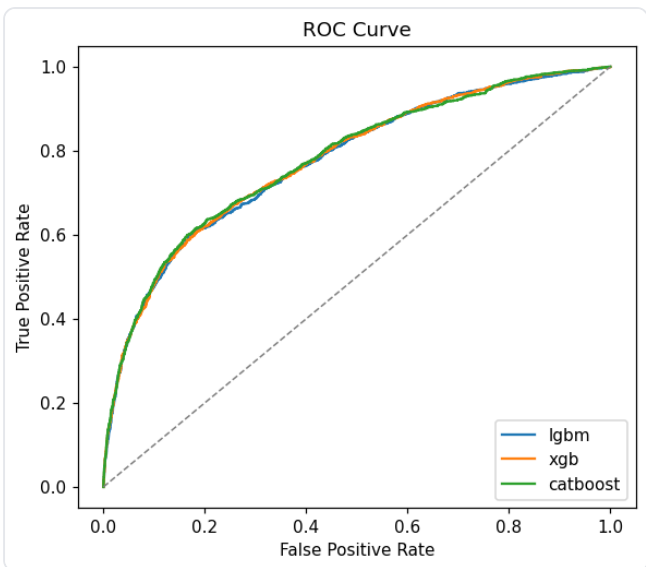
model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc	best_iter (avg)
lgbm	0.7748	0.4271	3.1424	0.7917	0.7939	0.0784	0.1427	0.5468	198
xgb	0.7773	0.4233	3.1575	0.8182	0.6720	0.3474	0.4580	0.5534	179
catboost BEST	0.7790	0.4340	3.1198	0.8178	0.6590	0.3655	0.4702	0.5594	601

2 CV (OOF) 비교

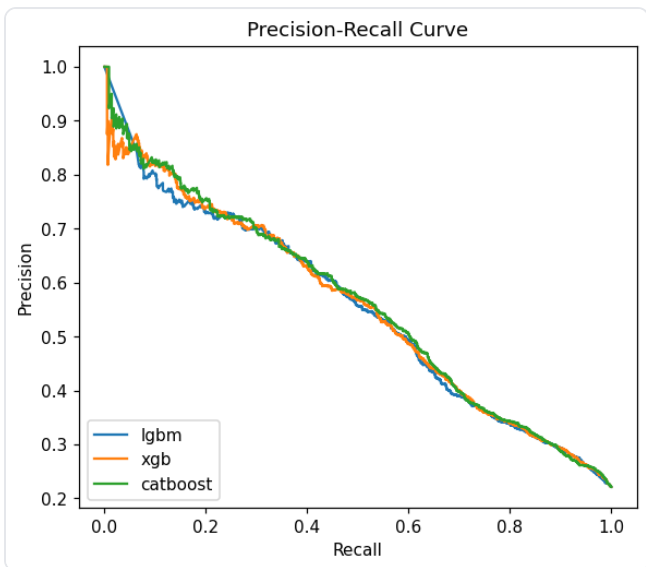
model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc
lgbm	0.7816	0.4327	3.1814	0.8201	0.6955	0.3321	0.4495	0.5548
xgb	0.7841	0.4349	3.1889	0.8215	0.6829	0.3601	0.4716	0.5580
catboost BEST	0.7839	0.4353	3.1946	0.8226	0.6790	0.3758	0.4838	0.5601

3 곡선 비교

ROC Curve



Precision-Recall Curve



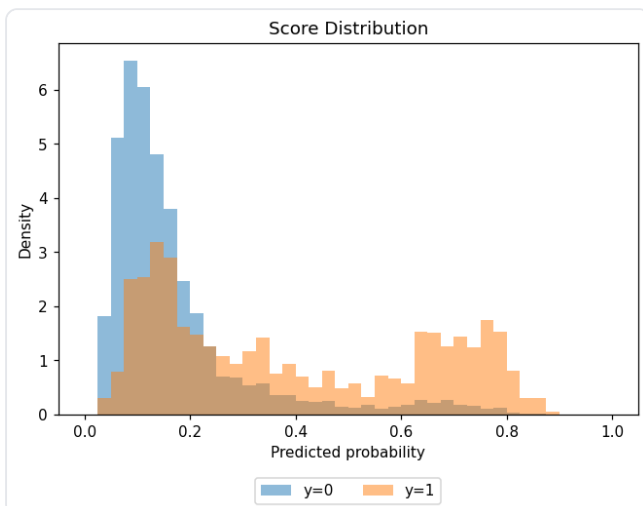
4 Best 모델 상세 — catboost

Feature Importance (top 30)

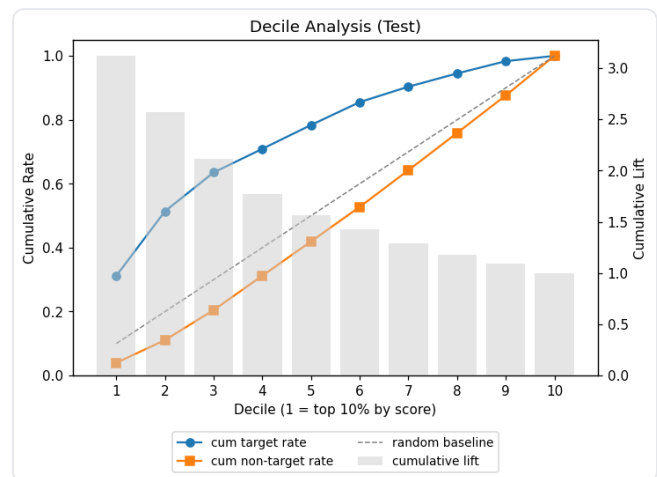
Rank	Feature	Importance	Rank	Feature	Importance
1	PAY_0	25.2403	16	PAY_AMT6	2.6288
2	LIMIT_BAL	9.8115	17	BILL_AMT6	2.5764
3	BILL_AMT1	8.1927	18	PAY_AMT5	2.5087
4	PAY_2	6.0274	19	SEX	1.5690
5	PAY_3	4.4471			
6	BILL_AMT2	4.3936			
7	PAY_AMT2	4.3723			
8	PAY_AMT3	4.2982			
9	PAY_4	4.2650			
10	PAY_AMT1	4.1239			
11	AGE	3.5923			
12	PAY_AMT4	3.0514			
13	BILL_AMT5	3.0093			
14	PAY_6	2.9744			
15	PAY_5	2.9178			

10-분위 분석 (테스트셋, 점수 내림차순) 및 점수 분포

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.595 ~ 0.886	600	414	186	414	186	600	10.0%
2	0.320 ~ 0.594	600	268	332	682	518	1200	20.0%
3	0.214 ~ 0.320	600	161	439	843	957	1800	30.0%
4	0.170 ~ 0.213	600	98	502	941	1459	2400	40.0%
5	0.144 ~ 0.170	600	99	501	1040	1960	3000	50.0%
6	0.122 ~ 0.144	600	95	505	1135	2465	3600	60.0%
7	0.103 ~ 0.122	600	64	536	1199	3001	4200	70.0%
8	0.086 ~ 0.103	600	55	545	1254	3546	4800	80.0%
9	0.067 ~ 0.086	600	51	549	1305	4095	5400	90.0%
10	0.026 ~ 0.067	600	22	578	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4422	251
True 1	842	485

5 학습 설정 요약

설정 키	값
target_column	default
categorical_columns	SEX, EDUCATION, MARRIAGE
id_columns	client_id
preprocessing.numeric_null_strategy	median
preprocessing.categorical_null_strategy	most_frequent
preprocessing.outlier_method	percentile
preprocessing.outlier_action	clip
preprocessing.skew_method	signed_log1p
preprocessing.skew_threshold	1.0
preprocessing.scaling_method	standard
training.cv_folds	4
training.early_stopping_rounds	50
training.primary_metric	roc_auc

설정 키	값
training.random_state	42
tuning.enabled	True
tuning.n_trials	15
tuning.cv_folds	3
tuning.timeout	None

6 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

Feature	Selection Frequency	채택
LIMIT_BAL	100.0%	✓ 채택
AGE	100.0%	✓ 채택
PAY_0	100.0%	✓ 채택
PAY_2	100.0%	✓ 채택
PAY_3	100.0%	✓ 채택
PAY_4	100.0%	✓ 채택
BILL_AMT1	100.0%	✓ 채택
PAY_AMT1	100.0%	✓ 채택
PAY_AMT2	100.0%	✓ 채택
PAY_AMT3	100.0%	✓ 채택
PAY_AMT6	100.0%	✓ 채택
PAY_5	98.8%	✓ 채택
BILL_AMT5	98.8%	✓ 채택
PAY_6	97.5%	✓ 채택
PAY_AMT4	96.2%	✓ 채택
PAY_AMT5	95.0%	✓ 채택
SEX	91.2%	✓ 채택
BILL_AMT6	88.8%	✓ 채택
BILL_AMT2	71.2%	✓ 채택
BILL_AMT3	56.2%	제외

Feature		Selection Frequency		채택
BILL_AMT4		28.7%		제외
EDUCATION		21.2%		제외
MARRIAGE		15.0%		제외